

Handling Categorical Values

Description

Categorical features must be converted to numerical format before model training. This project applies **feature mapping** and **label encoding** to transform any categorical columns.

Handling Categorical Values

Categorical features must be converted into numerical format before model training because machine learning algorithms work mainly with numbers.

Categorical data contains values in text or category form, such as gender, income type, education type, housing type, family status, weather condition, or status labels.

If these text values are passed directly to a machine learning model, the model may not understand them and may generate errors during training.

Therefore, categorical values are transformed into numerical values using encoding techniques.

In this project, feature mapping and label encoding are used to convert categorical columns into machine-readable format.

Feature Mapping

Feature mapping is used when categories have a clear order or fixed meaning.

In this method, each category is manually assigned a numerical value.

For example:

Low = 0

Medium = 1

High = 2

This makes the data easier for the model to understand.

Feature mapping is useful when the relationship between categories is known.

Label Encoding

Label encoding is another technique used to convert categorical values into numbers.

In label encoding, each unique category is assigned a different integer value.

For example:

Male = 0

Female = 1

Working = 0

Student = 1

Pensioner = 2

This technique is useful when categorical columns contain multiple unique values.

The **LabelEncoder** class from Scikit-learn can be used for this process.

After applying label encoding, all categorical columns are converted into numerical form.

This allows machine learning algorithms to process the dataset correctly.

Importance of Handling Categorical Values:

Handling categorical values is important because raw text data cannot be directly used for model training.

Encoding improves data compatibility with machine learning algorithms.

It helps convert meaningful categories into structured numerical values.

It also ensures that the dataset is fully prepared for training and testing.

If categorical values are not handled properly, the model may fail to train or may produce inaccurate results.

```
#independent features  
x=dataset.iloc[:,2:7].values  
  
#dependent feature  
y=dataset.iloc[:,9:].values
```